CouncilConnect An Artificial Intelligence Augmented Framework for Campus Governance and Operations

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***Abstract*—Higher education institutions may maintain formal student councils while the workflows connecting students, representatives, and administrators remain fragmented across messaging applications, notices, and manual processes. This paper presents CouncilConnect, an artificial intelligence augmented framework that unifies verified student representation, moderated communication, campus events, grievances, and supporting operational services. The work is positioned as a design and early implementation study. The current development phase concentrates on the PostgreSQL and Supabase data foundation, institutional hierarchy, row level access control, and event upload workflow. The target architecture combines React Native mobile and React web clients, Spring Boot services, PostgreSQL, Supabase authentication and storage, server side artificial intelligence, and institutional workflow rules. Three artificial intelligence assisted workflows are specified for contact request categorization, grievance prioritization, and semantic lost and found matching. The main contribution is a security aware governance architecture in which institutional identity and authorization remain authoritative while artificial intelligence provides bounded decision support. The paper distinguishes implemented or actively developed components from planned modules and defines functional, security, AI, and performance evaluations for subsequent development. No unsupported claims about final accuracy, latency, scalability, or user satisfaction are made.**

***Index Terms*—campus governance, higher education, student council, artificial intelligence, grievance management, semantic matching, role-based access control, PostgreSQL**

I. Introduction

Student representatives form an important interface between students and institutional authorities. Positions such as General Secretary, branch representative, class representative, association president, and anti-ragging representative create a recognizable chain of responsibility, but a formal hierarchy does not by itself provide a reliable digital workflow. The CouncilConnect project documentation identifies recurring governance, communication, operational, and administrative problems: students may not know the appropriate representative, informal communication may leave little institutional trace, event and grievance workflows may be fragmented, and historical council information can disappear when terms end [1], [2]. In India, the University Grants Commission (UGC) Regulations on the Redressal of Grievances of Students, 2023 require higher education institutions to constitute student grievance redressal committees and to follow a defined redressal procedure [3]. A traceable, status-tracked grievance workflow is therefore relevant to institutional practice, and the workflow proposed here is intended to complement, not replace, these formal committees.

Digital transformation in higher education increasingly uses mobile and web services to improve access to institutional support [4]. However, a campus governance platform has requirements that differ from a generic student portal. It must represent institutional authority, distinguish verified from

unverified representatives, restrict communication according to policy, and preserve an auditable history. The proposed system therefore treats governance as a workflow rather than as a directory.

The central problem addressed in this paper is how to combine institutional identity, hierarchical authorization, student-representative communication, operational campus services, and artificial intelligence (AI)-assisted processing in one architecture without allowing AI to replace institutional authority. Table I compares CouncilConnect (CC), in its target design, with informal messaging, notice boards, and enterprise resource planning (ERP) systems; the comparison is qualitative and reflects the authors' requirements analysis [1] rather than a measured benchmark, and implementation status is given in Table III.

The paper makes four contributions. First, it defines a unified architecture for governance and campus operations. Second, it maps institutional hierarchy to server-side authorization and database-level Row Level Security (RLS). Third, it defines fail-soft AI workflows for categorization, prioritization, and semantic matching. Fourth, it establishes an implementation-status model and evaluation methodology that separates current evidence from planned functionality.

II. Related Work and Research Gap

Informal communication through personal messaging groups is accessible but provides weak identity assurance, workflow structure, and institutional traceability. Physical notice boards and static council pages can publish information but provide limited interaction, verification, or lifecycle management. Generic academic information systems typically prioritize records such as attendance, marks, and timetables rather than student governance and representative communication. Generic ticketing systems provide useful intake and escalation primitives but do not naturally encode council hierarchy or representative verification. These observations are consistent with the project requirements analysis and higher-education governance literature [1], [5].

From a technical perspective, role-based access control provides a mature foundation for mapping organizational roles to permissions [6]. RESTful services provide a suitable interface between mobile or web clients and modular backend services [7]. Relational database design provides the persistence model needed to connect identities, council profiles, requests, events, and historical records [8]. For AI-assisted text processing, transformer architectures have established strong general-purpose language modeling capabilities [9]–[11], while sentence embeddings provide a practical basis for semantic similarity [12]. Recent authorization surveys review access control models, including role-based and attribute-based approaches [13], which is relevant because branch-scoped council authority depends on both a role and an attribute (the branch). Recent embedding work, including contrastive sentence encoders [14], text-to-text sentence encoders [15], the Massive Text Embedding Benchmark (MTEB) [16], and surveys of embeddings based on large language models (LLMs) [17], provides candidate encoders for semantic matching. On the security side, the Open Worldwide Application Security Project (OWASP) Top 10 for LLM Applications lists prompt injection and improper output handling among the principal risks of LLM-backed services [18], and surveys of hallucination in language generation [19] motivate validating model outputs rather than trusting them.

TABLE I
Capability Comparison (CC = CouncilConnect, target design; ERP = enterprise resource planning)

Capability	Informal	Notice	ERP	CC
Verified council directory	No	No	No	Yes
Role-aware authorization	No	No	Partial	Yes
Moderated communication	No	No	No	Yes
AI-assisted workflows	No	No	No	Yes
Grievance tracking	No	No	Partial	Yes
Event management	No	No	Yes	Yes
Analytics and archive	No	No	Partial	Yes

The research gap is therefore not the absence of individual technologies. The gap lies in combining verified student governance, moderated communication, institutional hierarchy, operational campus workflows, and bounded AI assistance in a security-aware platform. CouncilConnect addresses this integration problem for the institutional context described in the project materials.

III. System Model and Architecture

A. Governance Model

CouncilConnect uses three principal roles: Student, Representative, and Administrator. Students browse representatives, submit requests, register for events, and use operational services. Representatives manage approved communication and authorized events or grievances. Administrators verify representatives, manage escalations, and access protected institutional analytics.

The verification chain mirrors the project hierarchy: the head of department (HOD) or faculty administration verifies General Secretaries, and General Secretaries verify authorized branch representatives. Representative roles cannot be self-assigned. This hierarchy is enforced in application logic and database policies rather than being treated as a visual convention.

B. Functional Scope

The target platform contains governance and communication, event management, hostel grievance management, lost and found, smart campus navigation, analytics, and institutional archive modules. The conference paper focuses on the first three as the principal research contribution and treats the remaining services as supporting extensions of the same identity, data, and security foundation.

C. Architecture

The system separates client interfaces, backend workflow logic, persistent data, and external intelligence or infrastructure services. Representational state transfer (REST) application programming interfaces (APIs) provide a stable service boundary, while authentication, business authorization, and RLS provide defense in depth. Fig. 1 summarizes the layered architecture.

IV. Governance and Secure Workflows

A. Moderated Communication

The key governance workflow is a contact request. A student first locates a representative and submits a reason and description. The request is persisted with a pending state, categorized by the AI service, and placed in the representative's inbox. The representative then approves or declines the request. Approval unlocks a scoped direct-message (DM) thread; decline closes the request. This design converts an informal message into an auditable institutional object. Fig. 2 shows the resulting workflow.

B. Event and Grievance Workflows

Verified representatives can create events with title, venue, date, time, poster, and scope. Students can register and receive event updates, while participation data becomes available to authorized administrators. The current implementation effort treats event upload as the first operational workflow because it exercises authentication, authorization, validation, persistence, storage, and user-visible state.

The grievance module extends the same pattern to complaints. A student submits a complaint with a description and optional evidence; the intended workflow adds category, priority, responsible authority, status tracking, and escalation. These capabilities are specified in the project design but are not represented here as completed experimental results.

C. Security Model

Authentication uses institutional accounts with JSON Web Token (JWT)-based sessions [20]. Authorization is layered: client routing provides role-aware navigation, Spring Boot applies business rules, server-side verification functions enforce institutional hierarchy, and PostgreSQL RLS restricts data access. This defense-in-depth approach follows the principle that a UI restriction alone cannot protect sensitive data. Broken access control is also a major web application risk [21].

The external AI credential is never embedded in the client. AI calls originate from server-side execution. Message access is scoped to approved requests, and verification records retain the verifier and timestamp. The project materials also describe an alternative module-specific institutional-email access model [22]; the primary paper architecture retains role- and branch-aware authorization because it maps more directly to the council hierarchy and RLS policy model. Because branch-scoped rights depend on both a role and an attribute (the branch), this model can be read as role-based access control extended with attribute conditions, in line with the role-based and attribute-based approaches compared in recent authorization surveys [13].

TABLE II
Security Requirements and Enforcement

Requirement	Primary enforcement
Prevent role self-assignment	Invite-controlled onboarding and server checks
Restrict branch authority	Hierarchical verification and RLS policies
Protect private communication	Approved-request scope for message access
Protect AI credentials	Server-side AI execution only
Preserve auditability	Verifier identity, timestamps, and immutable archive snapshots
Contain AI failure	Persist core transaction before inference
Resist prompt injection	Text treated as data, closed-label validation, no extra model privileges

D. Threat Model and Security Requirements

The design considers five practical threat classes. First, a malicious client may attempt privilege escalation by changing a role value or calling an endpoint directly. Second, an authenticated student or representative may attempt to read records outside the permitted branch or workflow scope. Third, application credentials may be exposed if external AI secrets are shipped to a mobile or web client. Fourth, an external AI service may return an invalid result or become unavailable. Fifth, student-authored free text is untrusted input to the language model and may contain instructions intended to steer the assigned label (prompt injection). The architecture addresses these threats through server-side authorization, RLS, server-side secret handling, closed-label output validation (1), and fail-soft AI processing. The design treats request text strictly as data, gives the model no privilege beyond proposing a category, and rejects any output outside the allowed label set, consistent with OWASP guidance on prompt injection, improper output handling, and excessive agency [18]. Table II maps each requirement to its enforcement point.

This model follows the broader security principle that authorization should be enforced as close to the protected resource as practical. Frontend checks improve usability but are not considered authoritative. Backend checks express workflow semantics, while RLS provides a final data-access boundary.

V. Artificial Intelligence Methodology

AI is deliberately constrained to workflow assistance. It does not appoint representatives, approve communication, resolve complaints, or make disciplinary decisions. The current design specifies three uses: contact-request categorization, grievance categorization and priority assistance, and semantic lost-and-found matching.

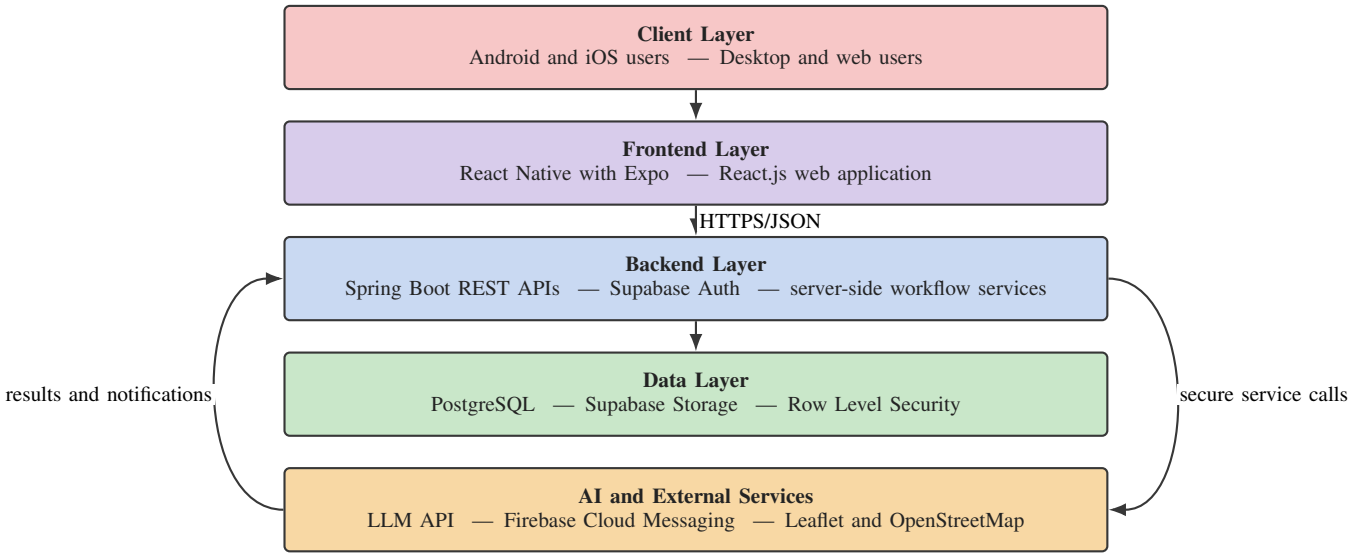


Fig. 1. Condensed architecture of CouncilConnect. The same identity and data foundation supports governance, operational workflows, and server-side AI services.

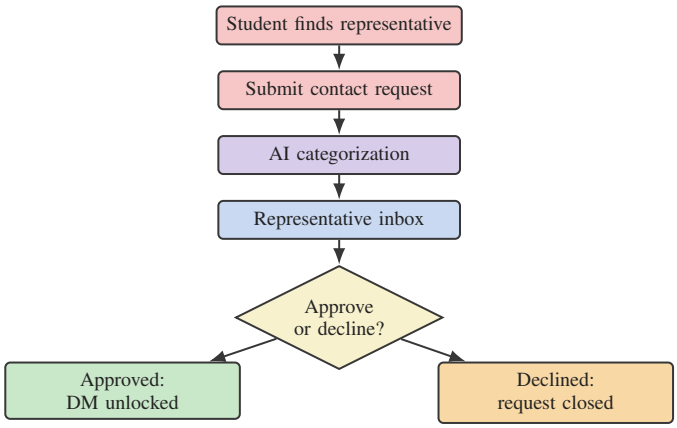


Fig. 2. Moderated contact-request workflow. AI supplies a category while the representative retains approval authority.

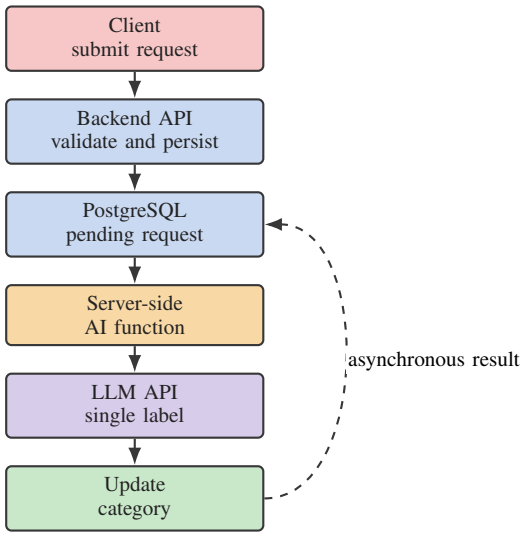


Fig. 3. Server-side AI categorization with fail-soft persistence.

A. Contact Request Categorization

The request title and description are sent from the backend to an LLM service through a protected server-side function. The output is constrained to four labels: Academic, Personal, Event, or Urgent. The initial database write does not depend on model availability. If the model fails or returns an invalid label, the request remains stored without a category. This fail-soft boundary prevents a third-party AI outage from becoming a user-facing transaction failure. For a request r with raw model output $g(r)$, the stored category is

$$\hat{c}(r) = \begin{cases} g(r), & g(r) \in \mathcal{L}, \\ \varnothing, & \text{otherwise}, \end{cases} \tag{1}$$

where $\mathcal{L} = \{\text{Academic, Personal, Event, Urgent}\}$ and $\varnothing$ denotes an unclassified request that remains fully usable. Fig. 3 shows the persistence order.

B. Grievance Prioritization and Semantic Matching

For grievances, AI is intended to support category and priority estimation, with human or institutional workflow rules remaining authoritative. For lost and found, item descriptions and contextual fields such as location can be embedded and compared. Sentence-BERT provides a practical basis for semantic similarity [12]. The similarity threshold must be calibrated using labeled campus reports rather than selected arbitrarily. Let $\phi(\cdot)$ denote a sentence encoder, $\mathbf{e}_x = \phi(x)$ the embedding of a report text x, ℓ a lost-item report, and $\mathcal{F}$ the set of found-item reports. The similarity of a pair and the resulting candidate set are

$$s(\ell, f) = \frac{\mathbf{e}_\ell \cdot \mathbf{e}_f}{\|\mathbf{e}_\ell\| \, \|\mathbf{e}_f\|}, \tag{2}$$

$$\mathcal{M}(\ell) = \{\, f \in \mathcal{F} \;:\; s(\ell, f) \geq \tau \,\}, \tag{3}$$

where τ is the similarity threshold. Because benchmarks such as MTEB report that no single embedding method performs best on every task [16], candidate encoders [14], [15], [17] will be compared against Sentence-BERT on labeled campus pairs before τ is fixed.

C. AI Reliability and Evaluation

Transformer-based language models provide the technical basis for modern text processing [9]–[11], but model behavior can vary with input ambiguity and external service changes. Language models can also generate content that is unfaithful to their input [19], so any output outside $\mathcal{L}$ is treated as unclassified by (1). Evaluation will therefore use manually reviewed data. Classification will be measured with precision, recall, and macro-F1 (defined in (4) and (5)), while matching will be evaluated using labeled relevant and irrelevant candidate pairs. Invalid outputs, service failures, and ambiguous inputs will be recorded as explicit failure cases. For each class $c \in \mathcal{L}$, precision P_c, recall R_c, and the macro-averaged F1 score are

$$P_c = \frac{\mathrm{TP}_c}{\mathrm{TP}_c + \mathrm{FP}_c}, \qquad R_c = \frac{\mathrm{TP}_c}{\mathrm{TP}_c + \mathrm{FN}_c}, \tag{4}$$

$$\mathrm{F1}_{\mathrm{macro}} = \frac{1}{|\mathcal{L}|} \sum_{c \in \mathcal{L}} \frac{2 P_c R_c}{P_c + R_c}, \tag{5}$$

where TP_c, FP_c, and FN_c count true positives, false positives, and false negatives for class c. Matching quality will use the same precision and recall over labeled candidate pairs at threshold τ.

VI. Data Design and Authorization

The relational schema is centered on `users`, with council-specific information in `council_profiles`. Communication uses `contact_requests` and `messages`; event workflows use `events` and `event_registrations`; grievances use `grievances`; lost and found uses `lost_found_items`; and historical council records use `archive`. This structure follows the relational model and uses universally unique identifiers (UUIDs) for stable cross-service references [8]. Fig. 4 shows the condensed relational model.

The principal integrity constraints are foreign-key relationships, unique request conditions, approved-request requirements for messaging, verifier identity and timestamp retention, and read-only archival snapshots. RLS is applied at the database layer so that unauthorized clients cannot obtain protected records merely by bypassing frontend controls.

VII. Implementation Status and Evaluation

A. Implementation Pattern

The intended backend follows a controller-service-repository separation. Controllers validate request structure and authentication context, services apply institutional rules, repositories perform persistence operations, and integration components communicate with storage, notifications, and AI services. This separation is useful for CouncilConnect because the same authorization rules must be reused by mobile and web clients.

TABLE III
Implementation Status

Area	Status	Focus
PostgreSQL schema	Active	Entities, UUIDs, constraints
Hierarchy and RLS	Active	Verification and access policy
Event upload	Active	Authorized creation and poster storage
Governance communication	Planned	Requests and moderated DM
AI workflows	Planned	Categorization and prioritization
Grievance workflow	Planned	Status and escalation
Lost and found	Planned	Semantic matching
Navigation, analytics, archive	Planned	Supporting services

The active event-upload workflow provides the clearest current implementation path. An authenticated representative submits event metadata; the service verifies the representative's authority, validates required fields, creates the event record, stores the poster under an access policy, and returns the resulting event to the client. The sequence is deliberately reusable: authenticated actor, validated input, authorized operation, persistent state, optional external service, and user-visible result.

For governance, the same pattern will be extended to contact requests. A duplicate active request is rejected, the request is persisted before AI inference, and only an approved request can expose a message thread. This ordering makes database state authoritative and reduces dependence on client-side sequencing.

The paper explicitly separates the target architecture from the current development state. At the time of writing, PostgreSQL schema construction, institutional hierarchy and access-control design, and event upload are the main active implementation areas. Governance communication, AI categorization, grievance processing, lost-and-found matching, navigation, analytics, and archive behavior remain planned or subsequent implementation stages. Table III summarizes the status of each area.

The current evaluation is therefore development-stage validation rather than a final performance benchmark. Foundation tests cover database creation, foreign-key integrity, state constraints, authentication, role assignment, and RLS behavior. Negative authorization tests should verify that students cannot modify council records, representatives cannot verify accounts outside their jurisdiction, and unauthorized users cannot access protected events or historical records.

For the event workflow, the current validation sequence is: authenticated actor, validated form, authorization check,

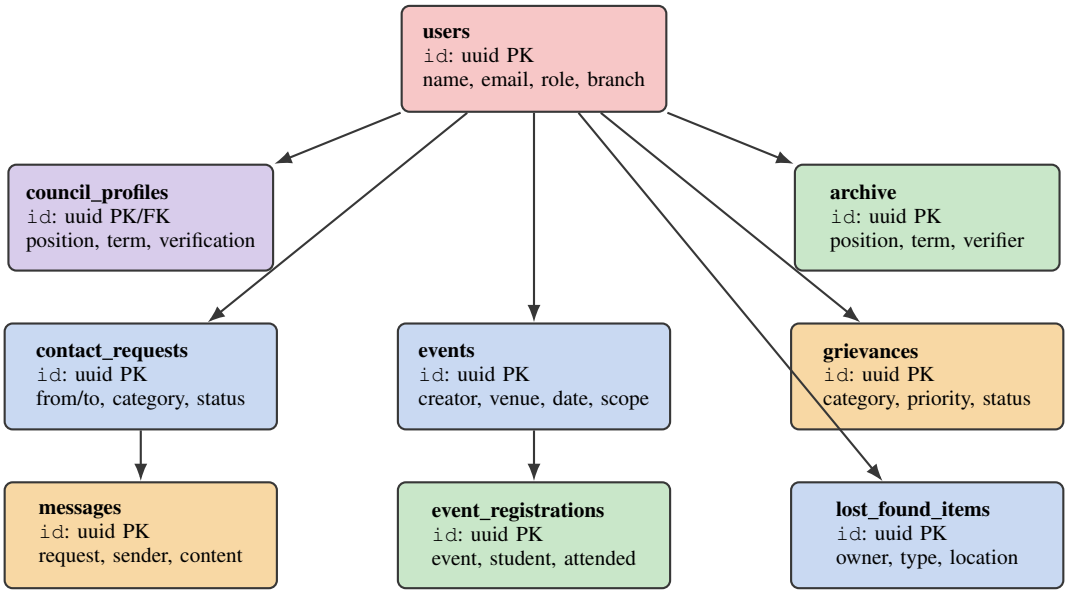


Fig. 4. Condensed relational model. Identity is the shared anchor for governance, communication, operations, and historical records.

database record, poster storage, and event listing. This sequence is intended to become the reusable acceptance pattern for subsequent modules.

Once sufficient labeled examples exist, AI evaluation will report classification precision, recall, and macro-F1 and will evaluate semantic matching using manually labeled candidate pairs. System experiments will later measure API latency, database latency, AI processing time, failure rate, concurrent load, and notification behavior. These measurements are intentionally not reported as completed results because the implementation is not yet at that stage.

B. Functional Acceptance Matrix

The planned acceptance criteria are intentionally observable at the system boundary. Foundation criteria require valid schema creation, foreign-key enforcement, correct role assignment, and rejection of unauthorized RLS operations. Event criteria require authorized event creation and poster association. Governance criteria require representative discovery, request persistence, AI categorization or safe unclassified fallback, and approval-controlled messaging. Subsequent module criteria require grievance lifecycle transitions, candidate lost-and-found matches, map representation, and immutable archive browsing. Table IV lists the planned criteria by stage.

Negative tests are particularly important for the governance model. Representative cases include a student attempting to update a council profile, a branch administrator attempting to verify a representative outside the branch, an unauthenticated client requesting protected records, and an attempt to modify an archived snapshot. Expected outcomes are authorization denial or rejection at the relevant layer. This provides evidence for the security contribution without requiring production-scale deployment.

TABLE IV
PLANNED ACCEPTANCE CRITERIA

Stage	Observable criterion
Foundation	UUIDs, foreign keys, states, authentication, and RLS are enforced.
Hierarchy	Each institutional account receives only its permitted verification and management actions.
Events	Authorized representative can create an event and associate poster storage correctly.
Governance	A valid request persists, and approval alone unlocks the DM workflow.
AI	Valid requests receive one allowed label or remain safely unclassified on failure.
Grievances	Complaints receive category, priority, status, and escalation.
Matching	Relevant lost/found candidates are surfaced above a calibrated similarity threshold.
Archive	Historical snapshots remain browsable and read-only after archival.

VIII. Discussion and Limitations

The principal design observation is that digital governance requires more than a directory. Identity, authority, request, response, and historical trace must form one workflow. The contact-request mechanism embodies this principle by converting informal outreach into a controlled institutional transaction.

AI is most useful in this setting as bounded decision support. A model can reduce manual categorization effort, surface priority signals, or identify candidate lost-and-found matches, while representatives and administrators retain authority over consequential actions. This separation also makes AI failures easier to contain.

Security is treated as a cross-layer property. Client restric-

tions, backend rules, server-side verification, and database RLS address different attack surfaces. This is particularly important for institutional applications because a valid login does not imply permission to perform every role-specific operation.

Several limitations remain. The project is an early implementation rather than a production deployment. Real-time communication is not yet the primary implementation path, and external LLM availability and behavior introduce dependency risk. The evaluation does not yet contain final AI benchmarks, usability results, or load measurements. The current design is institution-specific and would require tenant isolation, stronger operational monitoring, backup and disaster-recovery controls, and privacy governance for multi-institution deployment.

Future work includes real-time messaging, labeled in-house classifiers, predictive grievance escalation, recommendation and sentiment analysis, multi-tenant deployment, and streaming analytics. These are planned extensions rather than present capabilities.

IX. Conclusion

CouncilConnect presents a security-aware framework for digital campus governance that connects verified student representation, moderated communication, event workflows, grievance processing, and supporting campus services. Its central contribution is the integration of institutional hierarchy with layered authorization and bounded AI assistance.

The architecture separates client interfaces, backend workflow logic, relational persistence, and external intelligence services. PostgreSQL and RLS provide a durable security foundation, while server-side AI execution keeps model credentials outside the client and allows core transactions to survive inference failure. The current implementation concentrates on the database, authorization foundation, and event-upload workflow, with additional modules staged for subsequent validation.

The next phase should provide measured AI quality, security testing, usability evidence, and system performance results. Until those experiments are completed, CouncilConnect should be interpreted as an early implementation and system-design study rather than a fully validated production platform.

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